THE UNIVERSITY OF MELBOURNE

OzGrav
ARC CENTRE OF EXCELLENCE FOR GRAVITATIONAL WAVE DISCOVERY

OPEN ASTROPHYSICS CODEBASES

DEVELOPED AND MAINTAINED BY THE UNIVERSITY OF MELBOURNE GROUP

Audioband GWs (LVK)

Continuous-wave (CW) detection:
Hidden Markov model + Viterbi

Adaptive noise cancellation:
Recursive least squares

Nanohertz GWs (PTA)

CW parameter estimation for SMBH binaries:
Kalman filter

GW background detection:
Kalman filter

NEUTRON STAR THEORY

Quantum fluids

Vortex motion:
Gross-Pitaevskii solver
N-body solver

Thermal evolution model for hyperon stars

Neutron star oscillations:
Numerical relativity solver

Magnetic fields

Magnetic mountains:
Grad Shafranov solver
General relativistic formulation

PULSAR TIMING

Radio

Pulsar glitch detection
Hidden Markov model

Pulsar detection in compact binaries
Hidden Markov model

Two-component pulsar parameter estimation
Kalman filter

Modelling of young pulsar secular spin-down and timing-noise
Brownian Framework

X-ray

Low-mass X-ray binary parameter estimation
Kalman filter